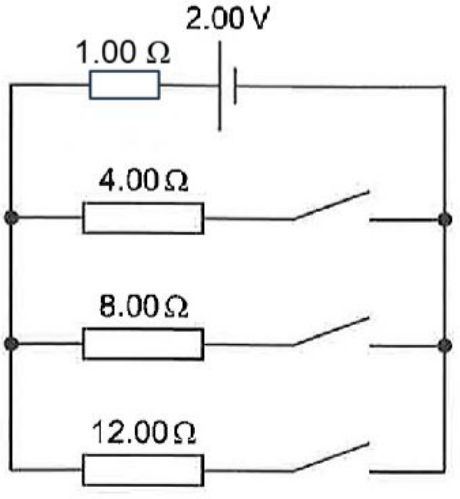
19 A cell of e.m.f. 2.00 V and internal resistance 1.00 Ω is connected to three resistors of 4.00 Ω, 8.00 Ω and 12.00 Ω as shown in the circuit diagram.



Ans: C

When every switch is closed, all three resistors are connected in parallel across the 2.00 V cell.

$$\frac{1}{R_{parallel}} = \frac{1}{4} + \frac{1}{8} + \frac{1}{12} = \frac{11}{24}$$

$$R_{parallel} = \frac{24}{11} = 2.182\ \Omega$$

The total resistance R_{total} becomes 24/11 + 1.00 = 35/11 Ω

$$I = \frac{V}{R_{total}} = \frac{2.00}{\frac{35}{11}} = \frac{2.00 \times 11}{35} = 0.629\ \text{A}$$

What is the current through the cell when every switch is closed?

A 0.0800 A **B** 0.500 A **C** 0.629 A **D** 0.917 A